

UTM

UNIVERSITI TEKNOLOGI MALAYSIA

SCSR2213 NETWORK COMMUNICATIONS

Lecturer:	**Dr. Adnan Khalid**	
Academic Session:	**2026-2027/2**	
Group Name	**NetCore**	
Assessment Title	**Task #1 - Project setup**	
Prepared by:	**Matric No.**	**Student name**
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Assessment mode:	**Group**	
Feedback from lecturer:		

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1.INTRODUCTION

This project is the first step in building Team Hub. Our goal is to design a modern and practical network for a two-floor facility in the Faculty of Computing. The building will include advanced computer labs and video meeting rooms, creating a fast, reliable, and future-ready environment.

Using tools like we created a smart floor plan that uses the floorplanner app while allowing easy maintenance and future expansion. We aim to build a strong network system that keeps data secure and provides high-speed performance for both students and staff. The design also focuses on creating a smooth user experience, where technology supports learning without complications.

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Floor Layout

The building is divided into two main levels:

- Ground Floor:

This floor is designed for collaboration and daily student activities. It includes a hybrid classroom, a video meeting room, and a large student lounge with study spaces and a copier. It also has prayer rooms for males and females, restrooms, and wide hallways for comfort and easy movement. The open layout helps create a welcoming and interactive environment for students.

Upper Floor:

This floor is the technical center of the building. It includes three labs: General Lab 1, General Lab 2, and the Embedded Lab. Each lab can support at least 30 workstations, making it suitable for both current and future needs. The layout ensures that students have enough space to work efficiently while using advanced equipment.

Design Approach

All labs are placed on the upper floor to make network setup and cable management simpler. This helps reduce complexity and makes maintenance easier for the technical team. It also allows better organization of network devices and improves overall system performance.

Each team member worked on specific tasks, such as designing parts of the floor plan and organizing the project details. Regular communication and teamwork helped us stay organized and maintain a clear direction throughout the project.

Conclusion

Our design focuses on simplicity, flexibility, and efficiency. It creates a comfortable and modern learning space for students and staff while also being ready to grow and adapt in the future. Overall, the project reflects a balanced combination of good planning, teamwork, and smart use of technology.

2.FLOOR DESIGN EXPLANATION :

The first floor features the faculty director’s office, a laboratory with space for 30 workstations, a conference room, and a break room.

The faculty dean’s office is positioned to have a clear view and easy access to the main lab and conference areas, allowing smooth coordination and supervision. The lab is designed for efficient use of space and network setup, while the conference room provides a dedicated area for meetings and collaborative activities. The break room offers a comfortable spot for relaxation and informal discussions.

This layout creates a functional and organized environment, balancing work, learning, and comfort, while leaving room for future expansion.

3.GROUND FLOOR DESIGN :

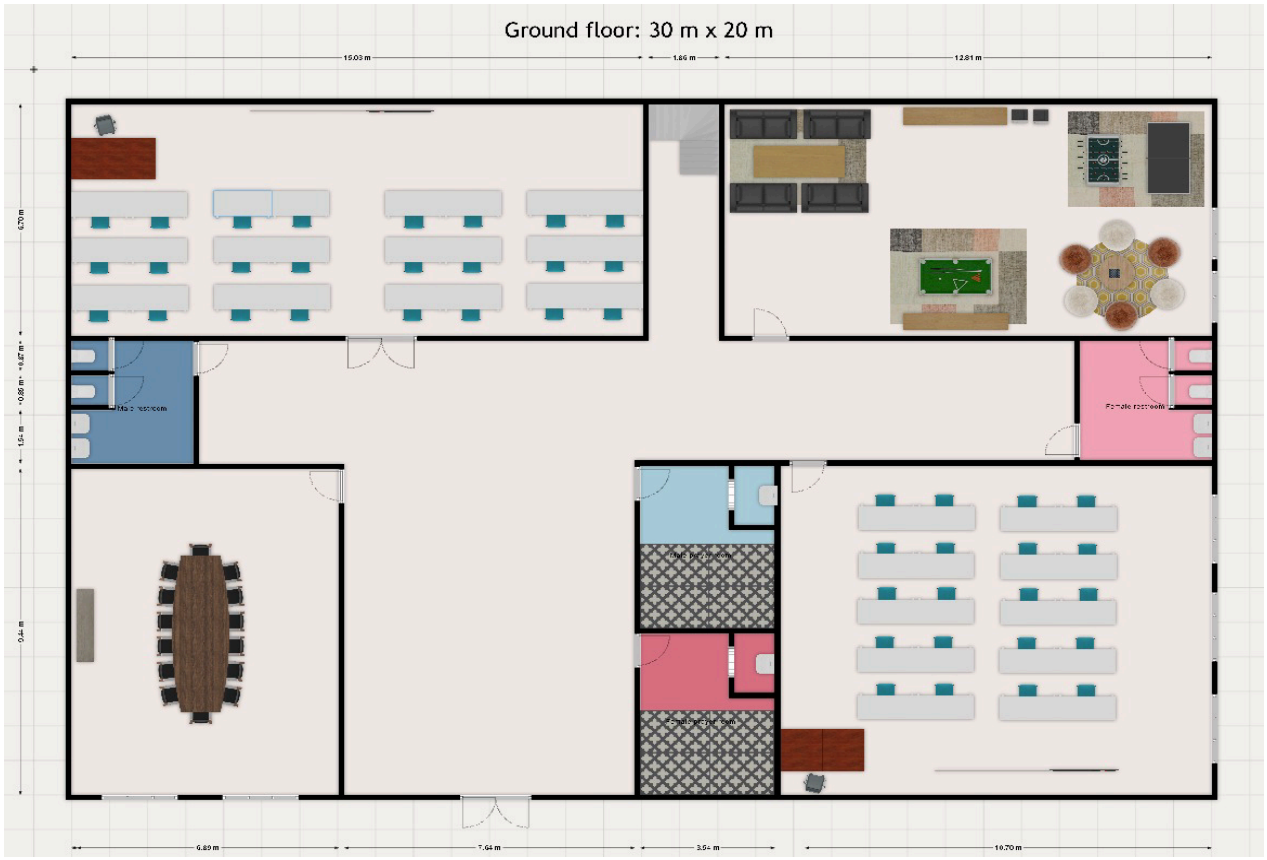


Figure 2.1 Ground-Level Floor Plan

The ground floor includes a hybrid classroom, a video conference room, a student lounge, as well as separate prayer rooms and bathrooms for males and females. This floor is designed to support both learning and collaboration, offering flexible spaces for teaching, discussions, and group activities.

The hybrid classroom and video conference room are placed in accessible locations to ensure smooth use and stable network connectivity. The student lounge provides an open and comfortable space where students can relax or study, with full wireless access available.

In addition, the prayer rooms and bathrooms are carefully positioned to ensure easy access and convenience for everyone. Overall, the layout creates a balanced environment that combines comfort, functionality, and efficiency.

4.FIRST FLOOR DESIGN:



figure 2.2 First Level Floor Plan

The Dean's Office is designed as a quiet and professional space that supports administrative work and decision-making. It is located in an accessible area to allow smooth communication with staff and easy supervision of the facility. The office provides a comfortable environment for meetings, planning, and daily tasks, while maintaining privacy and organization. Its placement within the floor layout helps ensure efficient coordination between academic and administrative activities.

5. Classroom:

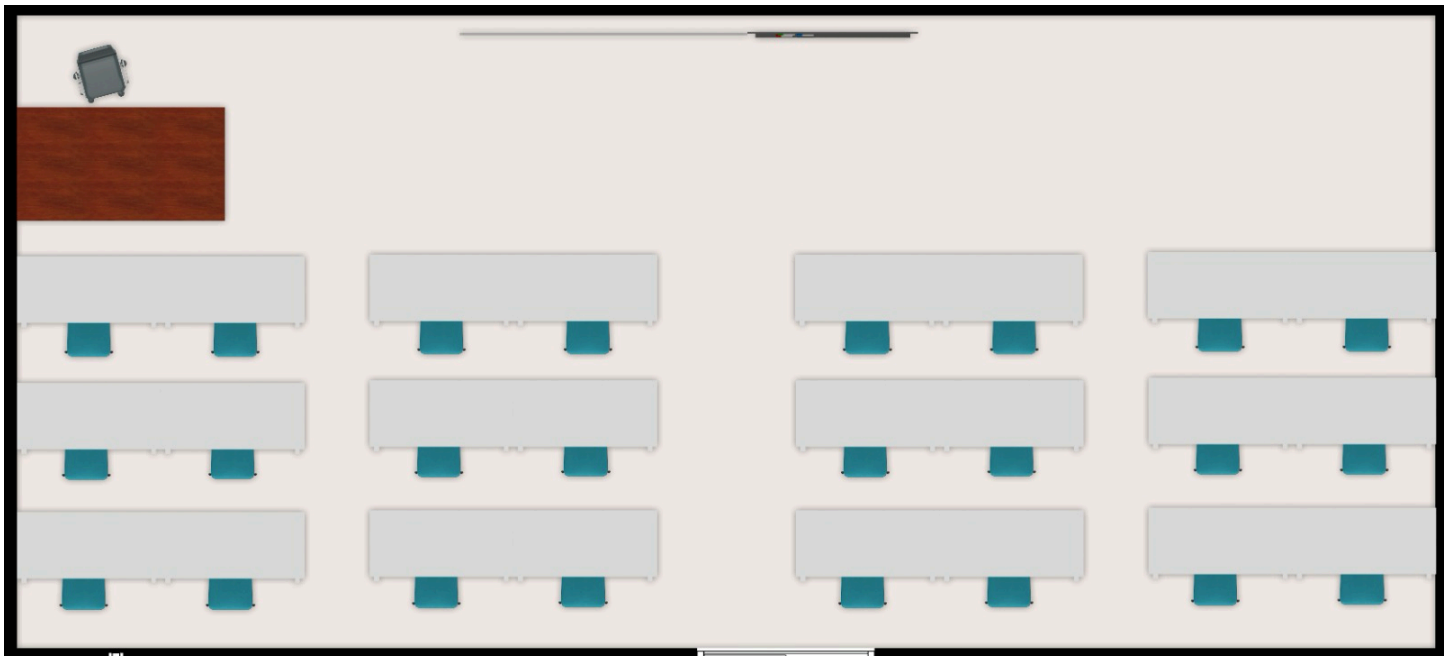


figure 2.3 Hybrid Classroom Floor Plan

The hybrid classroom supports both physical and virtual learning. It is equipped with video conferencing systems, interactive smart boards, wireless microphones, and high-speed internet connectivity. This setup allows the lecturer to conduct face-to-face sessions while simultaneously connecting with remote participants. The design ensures high audio-visual quality and comfortable seating arrangements to enhance the teaching and learning experience.

6.Video Conference Room:

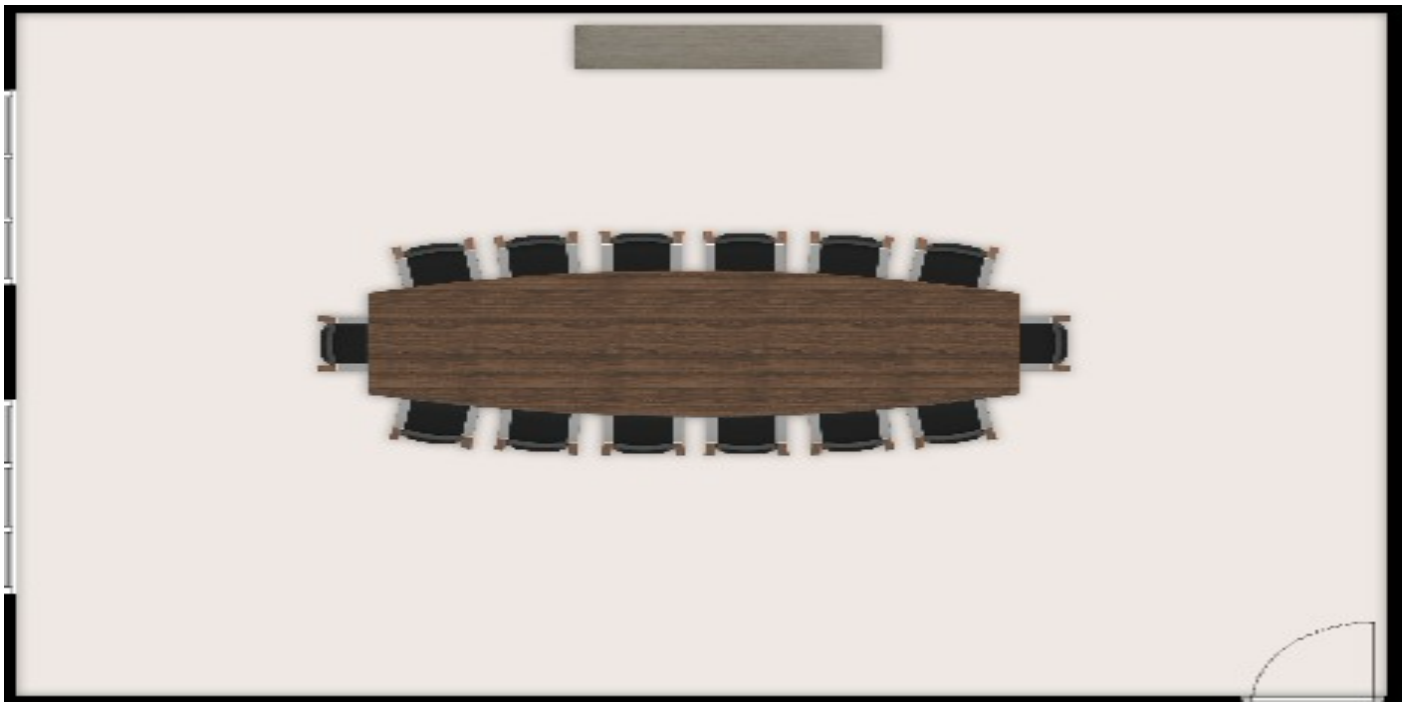


Figure 2.4 Video Conferencing Room Floor Plan

The video conference room is built for virtual meetings, presentations, and collaborative sessions. It includes high-definition cameras, conference microphones, a large display screen, and reliable network connections. The room supports communication between staff, students, and external partners, promoting remote project coordination and academic discussions.

7.Student Lounge:



figure 2.5 Student Lounge Floor Plan

The student lounge is thoughtfully designed with a range of amenities that support both relaxation and productivity. In one area, a cozy seating setup featuring four comfortable sofas and a large table creates a perfect environment for group discussions or individual study. For leisure and stress relief, the space offers recreational options such as a pool table, foosball, and a ping-pong table, allowing students to unwind through light physical activity. Furthermore, a circular seating space with vibrant cushions acts as a gathering point, encouraging social interaction and a sense of community among students. Overall, the lounge successfully combines academic support with opportunities for relaxation, providing a flexible and refreshing space between classes.

8.Cisco Lab:

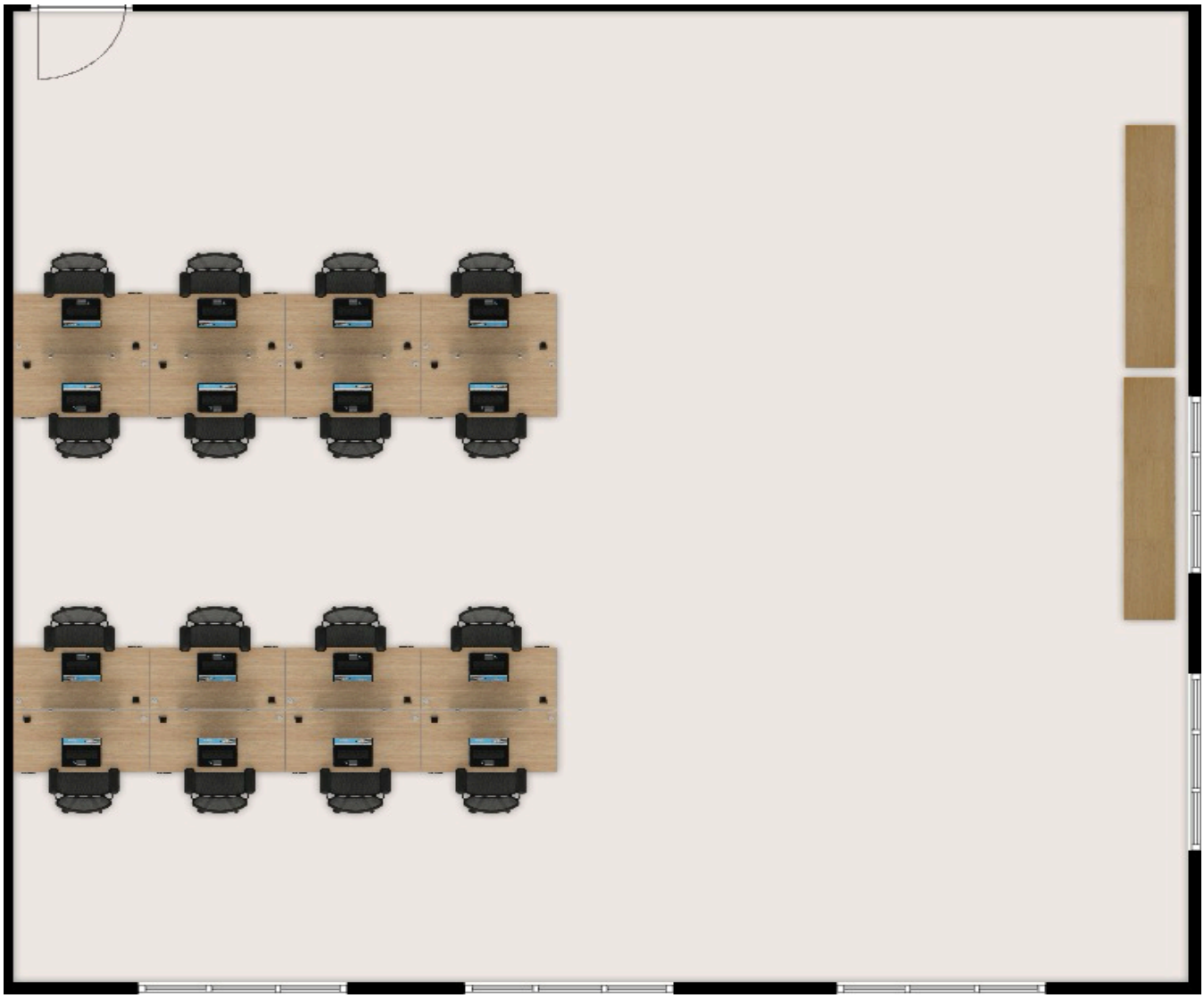


figure 2.6 Cisco Lab Floor Plan

The Cisco Lab is designed for network configuration, troubleshooting, and implementation training. It includes 36 computer workstations, 9 Cisco routers and switches, patch panels, and structured cable systems to support hands-on networking activities. The lab is equipped with high-speed wired connections and secure access points for real-time simulation and testing. This setup supports the faculty's goal of preparing students for professional networking certifications and 4IR-related technologies.

9.Embedded Lab:



Figure 2.7 Embedded Lab Floor Plan

The embedded lab focuses on IoT, digital systems, and sensor-based projects. It is equipped with development kits, microcontrollers, testing instruments, and computers connected through a stable local area network. The lab provides both wired and wireless connectivity to allow students to integrate hardware and software systems efficiently. The arrangement ensures enough workspace for hardware setup and prototype testing.

10. dean’s room:

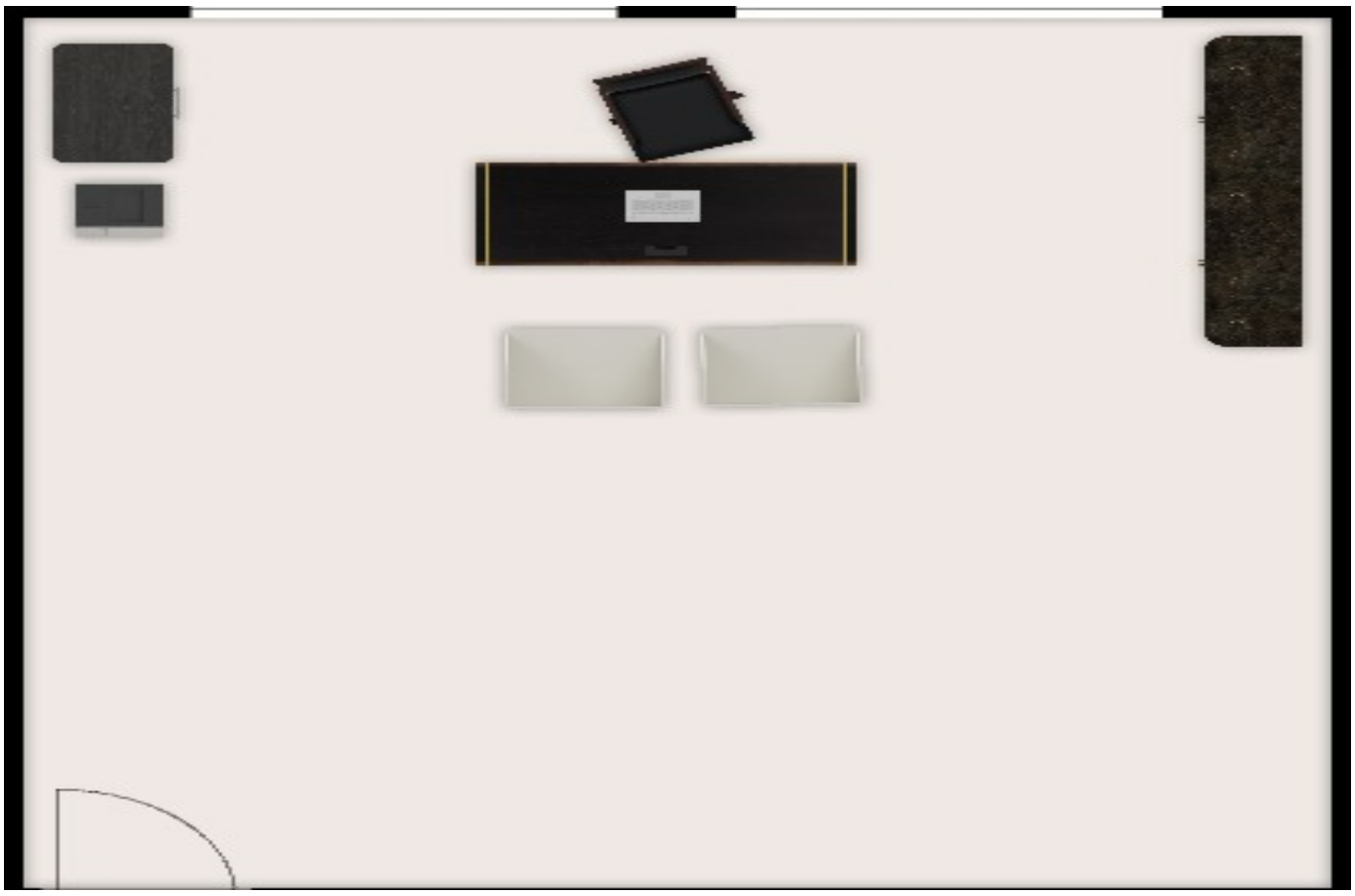


Figure 2.8 dean’s room Floor Plan

The Dean’s Office is designed as a quiet and professional space that supports administrative work and decision-making. It is located in an accessible area to allow smooth communication with staff and easy supervision of the facility. The office provides a comfortable environment for meetings, planning, and daily tasks, while maintaining privacy and organization. Its placement within the floor layout helps ensure efficient coordination between academic and administrative activities.

11.MEETING :

Date	**29/3/2026**
Location	UTM LIBRARY
Agenda	Discussion of project task 1 Discussion of group name

12..ATTENDANCE:

Name	Time	Reason for absence
Youssef elsayed	10:00AM	-
Qusai mahmoud	10:00 Am	-

12.MINUTES:

NO	ITEM DISCUSS	IDEA/SUGGESTION
1.	**TEAM NAME**	**Netcore team**
2.	Tools or software use to design the floor plan	The website “floorplanner app" was decided to be used after discussion.
3.	Floor plan design	**Youssef elsayed** And **Qusai mahmoud** designed the floor plan. **Qusai** volunteered to draw for the ground floor. **Youssef elsayed volunteered** to draw for the first floor.
4.	Directory of floor	**Youssef** volunteered to complete the directory of the floor.
5.	Next meeting	01/4/2026—to gather everything
6.	The meeting ended	2:00pm

13 . Projected Marks

TASK 1		
ITEM	**MARKS**	**EXPECTED MARKS**
Group name and team members		
Proposed floor plan		
Description based on the case study		
30 workstations and equipment planned for each lab		
Include a scale.		
Clear, easy to read, and well labeled		
Creativity		
TOTAL		